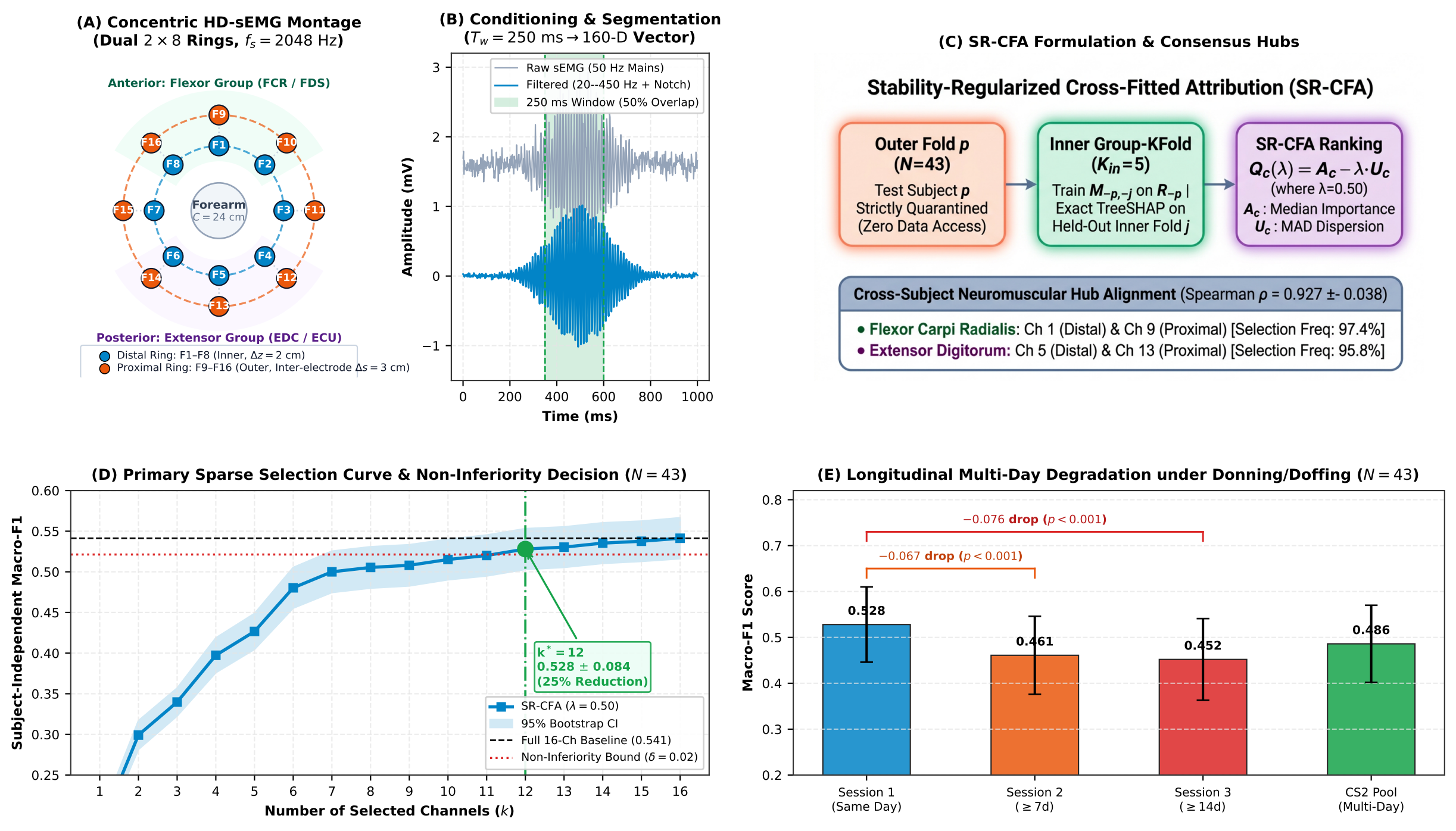

(A) Concentric HD-sEMG Montage
(Dual 2 × 8 Rings, $f_s = 2048$ Hz)
Anterior: Flexor Group (FCR / FDS)
F9
F16
F1
F10
F8
F2
Forearm
C = 24 cm
F15
F7
F3
F11
F6
F4
F14
F5
F12
F13
Posterior: Extensor Group (EDC / ECU)
Distal Ring: F1–F8 (Inner, Δz = 2 cm)
Proximal Ring: F9–F16 (Outer, Inter-electrode Δs = 3 cm)
(B) Conditioning & Segmentation
($T_w = 250$ ms → 160-D Vector)
Raw sEMG (50 Hz Mains)
Filtered (20--450 Hz + Notch)
250 ms Window (50% Overlap)
Amplitude (mV)
Time (ms)
(C) SR-CFA Formulation & Consensus Hubs
Stability-Regularized Cross-Fitted Attribution (SR-CFA)
Outer Fold p
(N=43)
Test Subject p
Strictly Quarantined
(Zero Data Access)
Inner Group-KFold
(K_{in}=5)
Train $M_{-p,-j}$ on R_{-p} |
Exact TreeSHAP on
Held-Out Inner Fold j
SR-CFA Ranking
$Q_c(\lambda) = A_c - \lambda \cdot U_c$
(where λ=0.50)
A_c : Median Importance
U_c : MAD Dispersion
Cross-Subject Neuromuscular Hub Alignment (Spearman ρ = 0.927 ±- 0.038)
• Flexor Carpi Radialis: Ch 1 (Distal) & Ch 9 (Proximal) [Selection Freq: 97.4%]
• Extensor Digitorum: Ch 5 (Distal) & Ch 13 (Proximal) [Selection Freq: 95.8%]
(D) Primary Sparse Selection Curve & Non-Inferiority Decision (N = 43)
Subject-Independent Macro-F1
Number of Selected Channels (k)
k* = 12
0.528 ± 0.084
(25% Reduction)
SR-CFA (λ = 0.50)
95% Bootstrap CI
Full 16-Ch Baseline (0.541)
Non-Inferiority Bound (δ = 0.02)
(E) Longitudinal Multi-Day Degradation under Donning/Doffing (N = 43)
Macro-F1 Score
−0.076 drop (p < 0.001)
−0.067 drop (p < 0.001)
0.528
0.461
0.452
0.486
Session 1
(Same Day)
Session 2
(≥ 7d)
Session 3
(≥ 14d)
CS2 Pool
(Multi-Day)